

The Kamke–Noncompactness Existence Theorem Extends to Every Caputo Order

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12 August 2026

Status. Candidate full positive answer to one of the explicit closing questions in arXiv:2511.01944. The source proves its existence theorem only for $0 < \alpha < 1$ and asks for an extension to arbitrary $\alpha > 0$. The theorem below supplies that extension, including all initial data and an explicit full-interval criterion.

1 Statement

Let E be a real Banach space, $\alpha > 0$, and $m = \lceil \alpha \rceil$ (with $m = \alpha$ when α is an integer). Given $u_0, \dots, u_{m-1} \in E$, put

$$P(t) = \sum_{j=0}^{m-1} \frac{(t-a)^j}{j!} u_j. \quad (1)$$

For this packet the arbitrary-order operators are defined by

$$D_{a+}^\alpha h = D^m J^{m-\alpha} h, \quad {}^C D_{a+}^\alpha u = D_{a+}^\alpha (u - P), \quad (1a)$$

where J^ρ is the Riemann–Liouville integral (and J^0 is the identity), whenever the right sides exist. At integer order this is the ordinary m th derivative; below order one it is exactly the source’s definition. Fix $\tilde{\delta}, \beta > 0$, set $\tilde{I} = [a, a + \tilde{\delta}]$, and consider the tube

$$R = \{(t, x) : t \in \tilde{I}, \|x - P(t)\| \leq \beta\}.$$

A sublinear measure of noncompactness μ is understood in exactly the axiomatic sense of the source, and it has the *singleton property* if $\mu(\{x\}) = 0$ for every $x \in E$.

We call a continuous $w_\alpha : \tilde{I} \times [0, \infty) \rightarrow [0, \infty)$ a Kamke function of order α when $w_\alpha(t, 0) = 0$ and zero is the unique nonnegative continuous function z satisfying

$$z(t) \leq \frac{1}{\Gamma(\alpha)} \int_a^t (t-s)^{\alpha-1} w_\alpha(s, z(s)) ds, \quad \frac{z(t)}{(t-a)^\alpha} \longrightarrow 0. \quad (2)$$

This is the source definition, now made for every positive order.

Theorem 1 (arbitrary-order extension). *Suppose $f : R \rightarrow E$ satisfies:*

1. *f is uniformly continuous on R ;*
2. *$\|f(t, x)\| \leq M$ on R for some $M > 0$;*
3. *for every $t \in \tilde{I}$ and every nonempty bounded $X \subseteq \overline{B}(P(t), \beta)$,*

$$\mu(f(t, X)) \leq w_\alpha(t, \mu(X)). \quad (3)$$

Then the Caputo problem

$${}^C D_{a+}^\alpha u(t) = f(t, u(t)), \quad u^{(j)}(a) = u_j \quad (0 \leq j < m) \quad (4)$$

has a solution on $I = [a, a + \delta]$, where

$$\delta = \min \left\{ \tilde{\delta}, \left(\frac{\beta \Gamma(\alpha + 1)}{M} \right)^{1/\alpha} \right\}. \quad (5)$$

The constructed u is continuous, the derivative in (1a) exists and is continuous, and all identities in (4) hold. It is obtained as a fixed point of the corresponding Volterra equation (12).

For $0 < \alpha < 1$, $m = 1$ and $P \equiv u_0$, so Theorem 1 is exactly the source theorem. The moving tube is necessary for higher order: the prescribed initial derivatives generate P before the nonlinear term is integrated.

Corollary 2 (whole prescribed interval). *Under the hypotheses of Theorem 1, if*

$$M\tilde{\delta}^\alpha \leq \beta \Gamma(\alpha + 1), \quad (6)$$

then the solution exists on all of $\tilde{I}$. In particular, this gives the full-interval extension asked about separately by the source whenever its boundedness and noncompactness hypotheses hold on an invariant tube satisfying (6).

2 Two kernel facts for every positive order

Write

$$(J^\alpha g)(t) = \frac{1}{\Gamma(\alpha)} \int_a^t (t-s)^{\alpha-1} g(s) ds. \quad (7)$$

Lemma 3 (uniform modulus). *If $\|g(t)\| \leq M$ on I , then*

$$\|J^\alpha g(t) - J^\alpha g(s)\| \leq L_\alpha |t-s|^q, \quad q = \min\{\alpha, 1\}, \quad (8)$$

where one can take

$$L_\alpha = \begin{cases} 2M/\Gamma(\alpha + 1), & 0 < \alpha < 1, \\ M, & \alpha = 1, \\ M\delta^{\alpha-1}/\Gamma(\alpha), & \alpha > 1. \end{cases} \quad (9)$$

Proof. For $0 < \alpha < 1$, split the two Volterra integrals at the smaller time and integrate the difference of the decreasing kernels; this gives the first line. At $\alpha = 1$ the assertion is immediate. For $\alpha > 1$, $J^\alpha g$ is differentiable with derivative $J^{\alpha-1}g$, whose norm is at most $M\delta^{\alpha-1}/\Gamma(\alpha)$. $\square$

Lemma 4 (noncompactness through the kernel). *Let $H \subset \mathcal{C}(I, E)$ be bounded and uniformly equicontinuous. For every $t \in I$,*

$$\mu \left(\left\{ \int_a^t (t-s)^{\alpha-1} h(s) ds : h \in H \right\} \right) \leq \int_a^t (t-s)^{\alpha-1} \mu(H(s)) ds. \quad (10)$$

This holds for every $\alpha > 0$.

Proof. The source proves (10) by approximating the Bochner integrals with common Riemann sums and using homogeneity and subadditivity of μ . The proof uses only that the scalar kernel is nonnegative and integrable. Both properties hold for $(t-s)^{\alpha-1}$ for every $\alpha > 0$; its being singular when $\alpha < 1$ is not otherwise used. Passing to the limit is justified by the common equicontinuity modulus, exactly as in the source. $\square$

For continuous g , the semigroup identity gives directly

$$D_{a+}^{\alpha} J^{\alpha} g = D^m J^{m-\alpha} J^{\alpha} g = D^m J^m g = g. \quad (11)$$

Moreover, the derivatives of $J^{\alpha} g$ through order $m-1$ vanish at a . Consequently every fixed point of

$$u(t) = P(t) + J^{\alpha}(f(\cdot, u(\cdot)))(t). \quad (12)$$

satisfies both the differential equation and the full initial jet in (4).

3 Proof of the theorem

The numerical choice (5) gives

$$\frac{M\delta^{\alpha}}{\Gamma(\alpha+1)} \leq \beta. \quad (13)$$

Let ω_P be the modulus of continuity of P on I , let q, L_{α} be as in Lemma 3, and define $X_0 \subset \mathcal{C}(I, E)$ by

$$X_0 = \{u : \|u(t) - P(t)\| \leq \beta, \quad u(a) = u_0, \\ \|u(t) - u(s)\| \leq \omega_P(|t-s|) + L_{\alpha}|t-s|^q\}. \quad (14)$$

It is nonempty, bounded, closed, convex, and uniformly equicontinuous. Define on X_0

$$(Fu)(t) = P(t) + J^{\alpha}(f(\cdot, u(\cdot)))(t). \quad (15)$$

Equations (8) and (13) show $F(X_0) \subseteq X_0$. Uniform continuity of f makes F continuous.

Set

$$X_{k+1} = \text{cl co } F(X_k), \quad v_k(t) = \mu(X_k(t)). \quad (16)$$

Then $X_{k+1} \subseteq X_k$. The common modulus in (14) and the standard evaluation estimate for a sublinear measure of noncompactness make $\{v_k\}$ uniformly equicontinuous. It is uniformly bounded and decreases pointwise, hence converges uniformly to a continuous $v_{\infty} \geq 0$.

Uniform continuity of f and the common modulus (14) ensure that $f(\cdot, X_k(\cdot))$ is uniformly equicontinuous. The singleton property removes the deterministic polynomial $P(t)$ from the measure estimate. Lemma 4, (3), and (16) give

$$v_{k+1}(t) \leq \frac{1}{\Gamma(\alpha)} \int_a^t (t-s)^{\alpha-1} w_{\alpha}(s, v_k(s)) ds. \quad (17)$$

Uniform convergence of v_k and uniform continuity of w_{α} on the compact range involved allow passage to the limit, so v_{∞} satisfies the first part of (2).

It remains to verify the small-time condition in (2); this is the only step where the higher-order initial data might conceivably cause trouble. Put

$$y(t) = P(t) + \frac{(t-a)^{\alpha}}{\Gamma(\alpha+1)} f(a, u_0). \quad (18)$$

For every $u \in X_0$,

$$\|Fu(t) - y(t)\| \leq \frac{(t-a)^\alpha}{\Gamma(\alpha+1)}\gamma(t), \quad (19)$$

where

$$\gamma(t) = \sup_{u \in X_0} \sup_{a \leq s \leq t} \|f(s, u(s)) - f(a, u_0)\| \longrightarrow 0. \quad (20)$$

Indeed, (14) gives $u(s) \rightarrow u_0$ uniformly over $u \in X_0$ as $s \downarrow a$, and then uniform continuity of f gives (20). Inequality (19) persists under closed convex hulls. Homogeneity, subadditivity, and the singleton property therefore yield

$$v_1(t) \leq \frac{(t-a)^\alpha \gamma(t)}{\Gamma(\alpha+1)} \mu(\overline{B}(0, 1)). \quad (21)$$

Since $0 \leq v_\infty \leq v_1$, the second part of (2) follows. Hence $v_\infty \equiv 0$.

The usual measure of noncompactness on $\mathcal{C}(I, E)$ combines the common equicontinuity modulus with $\sup_t \mu(X_k(t))$. Its value on X_k tends to zero. The intersection axiom consequently gives a nonempty relatively compact closed convex set

$$X_\infty = \bigcap_{k \geq 1} X_k, \quad F(X_\infty) \subseteq X_\infty. \quad (22)$$

Schauder's theorem supplies a fixed point of F in X_∞ . By (11)–(12) it solves (4). This proves Theorem 1; Corollary 2 is immediate from (5).

4 Checks, scope, and literature screen

The proof was checked separately in the three kernel regimes $0 < \alpha < 1$, $\alpha = 1$, and $\alpha > 1$. The only changes after order one are the polynomial initial jet and the Lipschitz, rather than fractional Hölder, modulus in Lemma 3. The decisive estimate (21) remains of exact order $(t-a)^\alpha$ because the whole initial polynomial is placed in the deterministic center $y(t)$.

On 12 August 2026 the run indexes and current arXiv/web records were searched for the source identifier, its title, arbitrary-order Caputo equations in Banach spaces, Kamke functions, and measure-of-noncompactness existence theorems. Related arbitrary-order, semigroup, Hilfer, weak-solution, and special-measure results were found, but no result matching this source's general sublinear-measure/Kamke theorem with its singleton hypothesis and arbitrary initial jet was located. This bounded search supports, but cannot guarantee, novelty.

References

- [1] D. Oberta, *On the existence of solutions of fractional differential equations in Banach spaces*, arXiv:2511.01944 (2025), especially Theorem 3.1 and the closing questions in Section 5.